Calibration service report

AI-assisted interpretation / Deterministic tolerance checks / Technician-reviewed

INSTRUMENT AND JOB CONTEXT

Client: Harbour Industrial Services (fictional) | Project: Pressure transmitter service — synthetic demonstration

Instrument: PT-204-DEMO | Type: Pressure transmitter

Technician: Demo technician | Service date: 2026-10-18

Engineering unit (common): bar | Measurement points: 5

TOLERANCE ASSESSMENT

As-found: 1/5 inside supplied tolerance (OUTSIDE). As-left: 5/5 inside supplied tolerance (WITHIN). Improved: 5; unchanged: 0; deteriorated: 0.

Acceptance basis: absolute measured-minus-reference error <= the positive per-point, technician-supplied tolerance. This is not an accredited uncertainty or conformance determination.

MEASUREMENTS AND COMPUTED DEVIATIONS

POINT P00 | Reference 0 bar | Tolerance +/- 0.05 bar
FOUND 0.11 bar | signed error 0.11 | OUTSIDE tolerance
LEFT 0.01 bar | signed error 0.01 | WITHIN tolerance

POINT P25 | Reference 2.5 bar | Tolerance +/- 0.05 bar
FOUND 2.59 bar | signed error 0.09 | OUTSIDE tolerance
LEFT 2.52 bar | signed error 0.02 | WITHIN tolerance

POINT P50 | Reference 5 bar | Tolerance +/- 0.05 bar
FOUND 5.06 bar | signed error 0.06 | OUTSIDE tolerance
LEFT 5.01 bar | signed error 0.01 | WITHIN tolerance

POINT P75 | Reference 7.5 bar | Tolerance +/- 0.05 bar
FOUND 7.52 bar | signed error 0.02 | WITHIN tolerance
LEFT 7.50 bar | signed error 0 | WITHIN tolerance

POINT P100 | Reference 10 bar | Tolerance +/- 0.05 bar
FOUND 9.92 bar | signed error -0.08 | OUTSIDE tolerance
LEFT 9.98 bar | signed error -0.02 | WITHIN tolerance

Each error is measurement minus reference; absolute error is compared to the per-point absolute tolerance. All values are printed as entered or calculated, without rounding. No uncertainty or traceability claim is made.

AI-ASSISTED SUMMARY (TECHNICIAN REVIEWED)

Pressure transmitter PT-204-DEMO underwent calibration testing across five points from 0 to 10 bar. The as-found condition was outside of the specified absolute tolerance of 0.05 bar for four out of five measurement points. Following adjustments, the as-left condition was within tolerance for all tested points. Technician-reported notes indicate the device was inspected during a simulated bench check and appeared stable after a notional span adjustment.

EVIDENCE-LINKED OBSERVATIONS

[P00, P25, P50, P100] As-found measurement errors exceeded the absolute tolerance of 0.05 bar at 0, 2.5, 5, and 10 bar points.

[P00, P25, P50, P75, P100] Technician-reported notes state the sensor appeared stable after a notional span adjustment during a simulated bench check.

RECOMMENDED TECHNICIAN CHECKS

[P00, P25, P50, P100] Verify as-left accuracy against tolerance requirements following the reported span adjustment.

ORIGINAL TECHNICIAN FIELD NOTES (UNVERIFIED)

Synthetic training dataset only. Device was inspected during a simulated bench check. Sensor appeared stable after a notional span adjustment. No actual instrument was used and no compliance claim is made.

PROVENANCE AND SIGN-OFF

Deterministic arithmetic: decimal.js (client preview and independently recalculated server-side). AI provider: Cloudflare Workers AI. Model: @cf/google/gemma-4-26b-a4b-it. AI narrative reviewed and acknowledged by the operator prior to export.

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IMPORTANT LIMITATIONS

SERVICE REPORT PREPARATION ONLY. Results are comparisons against technician-supplied absolute tolerances, not accredited calibration, compliance certification, measurement uncertainty analysis, or a safety-to-operate decision. Verify instrument traceability, units, methods and acceptance criteria independently. A qualified technician must review and approve the content.